# Signs of an Impending Volcanic eruption

1. **Ground deformation**- Ground deformation refers to surface changes on a volcano, such as subsidence (sinking), tilting, or bulge formation, due to the movement of magma below the surface. Deformation changes at a volcano, such as those related to magnitude or location, may indicate that an eruption is about to occur.




Figure 4.1. Deformation of ground was noticeably seen in Eastern Turkey which was a possible effect of an earthquake.

2. **Seismic activity**—from earthquakes to swarms of earthquakes. Most unrest in volcanoes start with volcanic earthquakes. Volcanoes and earthquakes go hand in hand. The challenge is to determine which patterns of seismic waves precede an eruption. When magma makes its way up to the surface it breaks the rocks along the way, and in so doing, earthquakes are generated.

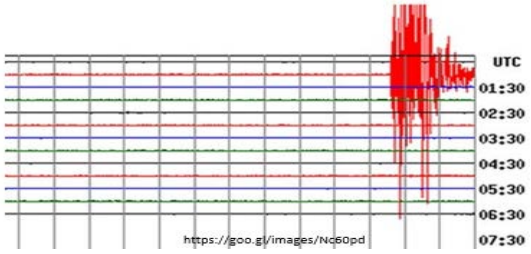



Figure 4.2. Shows a seismograph which records the seismic activity in the possible occurrence of an earthquake or volcanic eruption

3. **Gasses**—types and rate of emission. Gasses rise through vents called fumaroles (from the Latin for "smoke") and other cracks. Sometimes the concentrations are high enough to create acid rain that kills vegetation scientists have several ways to measure the rate of emissions more precisely. Scientists mount/ install infrared and correlation spectrometers from airplanes, for example, and fly through a plume of gas. These instruments read energy signatures—thermal output or electromagnetic frequencies—to identify and quantify the gasses.




Figure 4.3 Visible exhaustion of gas from Mount Sinabung Indonesia.

4. **Visual observations** – People living near volcanoes may observe premonitory events before an eruption. The following changes may be observed prior to the explosion of a volcano: Intensified steaming activity
    a. What used to be white steam slowly or drastically change to gray to dark (suggests increasing presence of ash).
    b. Drying up of vegetation, drying up of streams, water wells.
    c. Crater glow at the summit area.
    d. Increasing frequency of rolling rocks from the summit; localized landslides not related to heavy rains.
    e. Summit area appears to glow or becomes incandescent at night.

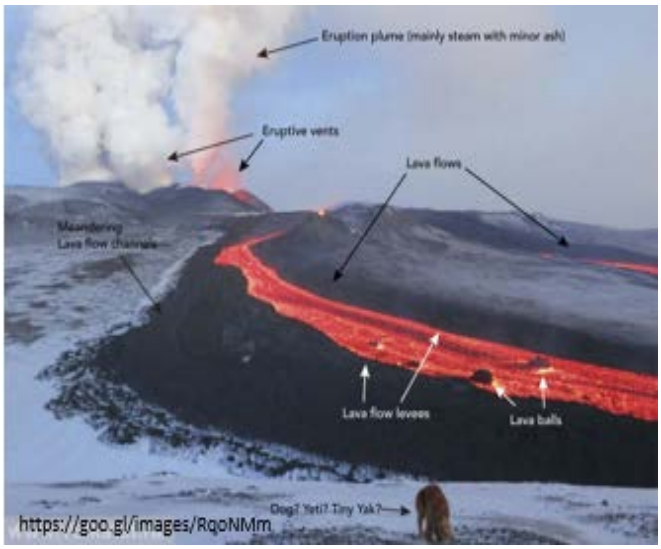



Figure 4.4. Visual illustration of an impending volcanic eruption in Tolbachik Russia.

5. Other sensory observations - This includes sense of smell, hear, and feel. People who are residing or at near a volcano can observe signs of an impending eruption through the following senses:
    a. Auditory observations - Hear rumbling sound
    b. Olfactory - Smell of sulfur (rotten eggs)
    c. Feel (Tactile) - Ground movement/ volcanic earthquake increasingly become felt

**References:**

CHED. (n.d.). *Different types of volcanic hazards*. Retrieved from http://teachtogether.chedk12.com/teaching_guides/view/337 on March 8, 2017.

EERI Admin. (n.d). *Ground deformation*. Retrieved from http://www.eqclearinghouse.org/2011-10-23-eastern-turkey/2011/10/30/photos-of-observed-ground-deformation/ on March 8, 2017.